



Revised Empirical Relations Between Earthquake Source and Rupture Parameters by Regression and Machine Learning Algorithms

SUKANTA MALAKAR,¹ ABHISHEK K. RAI,¹ VIJAY K. KANNAUJIYA,¹ and ARUN K. GUPTA²

Abstract—In this study, we have developed new empirical relations between various source and rupture parameters such as moment magnitude (M), surface rupture length (SRL), subsurface rupture length (RLD), rupture width (RW), rupture area (RA), and average (AD) and maximum slip (MD), based on an extensive database. The study involves about 476 global earthquakes that occurred between 1857 and 2023, covering a range of magnitudes (≥ 4.5) and faulting styles. The results indicate that relations between M-SRL, M-RLD, M-RW, M-RA, M-AD and M-MD correlate well for all types of faulting compared with previous studies. However, log-linear regression may not account for the nonlinear behaviour of rupture parameters, and these equations are separately used for each fault parameter, which leads to inconsistency in magnitude prediction. Hence, machine learning technique has been used to estimate earthquake magnitudes using various fault parameters simultaneously, which ensures consistency. In this study, we have employed an artificial neural network (ANN) and gradient-boosting machine regression (GBM) and examined their performance and applicability. Our analysis shows that gradient-boosting machine learning estimates earthquake magnitude better than regression equations, but the artificial neural network outperforms both. The result of this study would be beneficial for paleoseismic studies where reliable estimates of earthquake magnitudes and other source parameters are often difficult to estimate.

Keywords: Earthquake source parameters, Empirical relations, Regression analysis, Machine learning, ANN, GBM.

1. Introduction

Most of the destructive earthquakes are associated with surface ruptures threatening human life and infrastructures (Ghassemi, 2016). Infrastructure safety, especially in earthquake-prone regions,

requires structurally stable and strong constructions following approved designs for earthquakes of relatively large magnitudes (Nowroozi, 1985). Therefore, understanding and predicting active fault rupture geometries and kinematics are essential before formulating infrastructures and building design codes. Furthermore, fault-rupture scaling relationships and empirical formulas relating various fault parameters and magnitude are also used to estimate the maximum potential magnitude from known fault parameters (Nowroozi, 1985), which are often used in determining the probabilistic and deterministic seismic hazard in a given area of study. Thus, the relationships between various parameters have significantly critical applications in geotechnical, seismotectonics, seismic hazard assessment and paleoseismological studies (Konstantinou et al., 2005).

Many researchers have used statistical analysis to develop the empirical relations between earthquake magnitude and fault parameters (Ohnaka, 1978; Acharya, 1979; Slemmons, 1982; Bonilla et al., 1984; Slemmons et al., 1989; Wells & Coppersmith, 1994; Vakov, 1996; Mai & Beroza, 2000; Fujii & Matsu'ura, 2000; Stirling et al., 2002; Haiyun & Xiaxin, 2003; Wesnousky, 2008; Blaser et al., 2010; Shaw, 2013; Murotani et al., 2015; Thingbaijam et al., 2017). However, empirical relationships derived in these studies differ to some degree, primarily due to variations in the numbers and types of earthquakes used, and also due to difference in quality control parameters used for data selection. Many researchers have used the ordinary least square regression method to estimate reliable empirical relationships between moment magnitude and other fault parameters (Haiyun & Xiaxin, 2003). The relationships proposed

¹ Centre for Ocean, River, Atmosphere and Land Sciences (CORAL), Indian Institute of Technology Kharagpur, Kharagpur, West Bengal 721302, India. E-mail: abhishek@coral.iitkgp.ac.in

² Seismology Division, Ministry of Earth Sciences, New Delhi, India.

by Wells and Coppersmith (1994) are widely used to predict fault dimensions and slip parameters for earthquakes (Asim et al., 2019; Ghassemi, 2016; Haiyun & Xiabin, 2003). The relationship between various parameters can be further updated using the recent availability of updated high-quality catalogues of recent and paleo-earthquakes.

A homogeneous, complete and consistent earthquake catalogue is essential for properly understanding tectonic phenomena in a region (Das & Meneses, 2021). Including recent source parameters and rupture data derived from digital seismograms, InSAR analysis can significantly improve the empirical relationships between various earthquake parameters. Furthermore, widely used empirical relationships may not be suitable for earthquakes in some specific tectonic settings, such as subduction zones (Atkinson & Boore, 2003). Moreover, the conventional methodologies used to derive empirical relations based on regression equations may not account for the nonlinear behaviour of rupture parameters (Asim et al., 2019). These regression equations can predict different magnitude values using a specific fault parameter and another value for a different fault parameter. The observed minor deviations in derived magnitudes are considered random variations and within the error limit. Machine learning can incorporate the nonlinearity of earthquake source parameters, and can significantly improve relationships between various parameters used for hazard estimations (Asim et al., 2019).

Machine learning is a branch of computer science that identifies and extracts hidden meaningful relationships within the data that can be used to predict unknown data values (Reyes et al., 2013). Various machine-learning techniques have recently been applied in the earth sciences to understand various aspects of seismotectonics (Dhanya & Raghukanth, 2018; Asim et al., 2018, 2019; Ren et al., 2020; Chanda & Somala, 2021; Szafranski & Duan, 2022; Malakar & Rai, 2022; Rai et al. 2023; Kim et al., 2023). Gomez & Kadri (2021, 2023) considered the rupture length, width, duration, average speed (thus displacement) and orientations using acoustic data from earthquakes in the Pacific and Indian Oceans and determined earthquake magnitudes and real-time fault parameters. They applied two machine-learning algorithms, support vector machines and a random forest classifier, to the acoustic signal properties to estimate slip types and

earthquake magnitudes. The result shows that both machine learning algorithms show similar accuracy (between 70 and 80%) for binary classification of the dataset, while for multi-classification, overall classification accuracy $> 60\%$ has been achieved in both models. However, the study primarily focuses on events that occurred in the oceanic region, and the dataset size is limited (Gomez & Kadri, 2021). Asim et al. (2019) attempted to estimate earthquake magnitude using fault length, width, area and maximum displacement (depending on availability) parameters using an artificial neural network (ANN). They used the historical data with a magnitude $M \geq 5$ collected from previous studies (Wells & Coppersmith, 1994) and the SRCMOD database (Mai & Thingbaijam, 2014). However, their analysis excluded parameters such as average displacement and surface rupture length. Surface rupture is associated with regional tectonics, in which the fault rupture plane intersects the ground surface and can be correlated with earthquake magnitude. Furthermore, they selected only events from the catalogue with fault length, width, area and maximum displacement information and neglected other events. The methodology does not include estimating missing datasets and further utilising them to predict the earthquake magnitude. In our study, the derived regression relationships that are based on a more homogeneous earthquake catalogue derived by including data from recent studies.

In summary, we have a) used an updated earthquake catalogue by incorporating additional data derived from recently published literature, b) revised the empirical relationships between magnitude and various fault parameters and compared them with previous studies, and c) predicted earthquake magnitude by applying an artificial neural network (ANN) and gradient boosting machine (GBM) using fault parameters besides examining their performance and applicability. In the subsequent sections, the data source, applied methodology, result and discussion are given in detail.

2. Data

In this study, we have used earthquake source and rupture parameters compiled from various online

catalogues, the US Geological Survey Open-File Report (Hayes et al., 2017), SRCMOD (Mai & Thingbaijam, 2014) and published literature (Wells & Coppersmith, 1994; Ma & Wu, 2001; Papadimitriou & Karakostas, 2006; Papadimitriou et al., 2007; Omira et al., 2015; Xu et al., 2017; Boncio et al., 2018; Atzori et al., 2019; Fan et al., 2019; Li et al., 2020; Xu et al., 2020; Guo et al., 2021; Romano et al., 2021; Jiang et al., 2022; Mulia et al., 2022; Xu et al., 2023; Li et al., 2023; Barbot et al., 2023). The data include fault parameters of shallow-focus, continental intra-plate and inter-plate earthquakes. We have also included events that occurred in subduction zones, oceanic and continental plate boundaries, and oceanic plates. The compiled catalogue of source parameters includes the event location, slip type, magnitude, seismic moment, surface and subsurface rupture length, rupture width, area, average and maximum displacement. A particular emphasis was given to the measured source parameters and types reported in the literature, including teleseismic inversion, geodetic modelling or rupture area based

on aftershock distribution. Accordingly, we examined the datasets using the following criteria: a) a magnitude M_w of ~ 4.5 and greater, and based on the evaluation criteria proposed by Wells and Coppersmith (1994), the information is classified as reliable or unreliable, b) when multiple rupture models are produced by the author(s) for the same event, we use the most recent version (Thingbaijam et al., 2017), and c) the 'expert opinion' method was used to identify the outliers in the data (Sims et al., 2008).

The preliminary datasets are then classified based on measurement types, and correct values are selected for further analysis. The primary database was thoroughly examined and reduced to 476 earthquakes that occurred between 1857 and 2023. The location of earthquakes used in this study is shown in Fig. 1. The figure also shows the estimated population density for 2020 (CIESIN, 2018), major faults (Styron and Pagani, 2020), major tectonic plates and their general direction of movement, epicentral location of earthquakes ($M \geq 4.5$) with slip types and location of great earthquakes. Regarding slip types, the catalogue

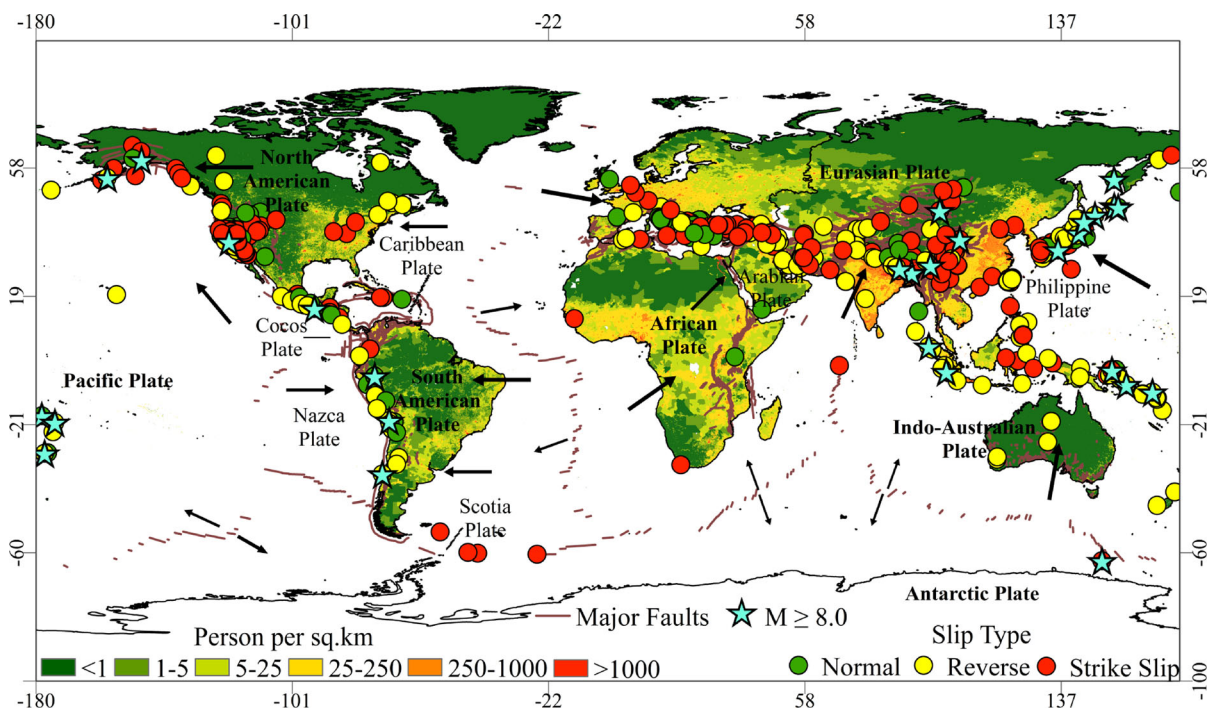


Figure 1

Locations of worldwide earthquakes ($M \geq 4.5$) that occurred between 1857 and 2023 and were used in this study. Earthquakes with different faulting styles are shown in different colours. Major tectonic plates, their average direction of movement and global population density are also shown for reference

includes 228 strike-slip earthquakes, 170 reverse, and 78 show normal faulting mechanisms. While categorising the catalogue based on magnitude, the number indicates that 122 earthquakes have a magnitude between 4.5 and 6.0, 185 events for magnitude 6.0–7.0, 141 events for magnitude 7.0–8.0 and ≥ 8.0 magnitude; the number of events was 28. The number of events shows that lower-magnitude earthquakes were not frequently reported during the pre-instrumental era.

The consolidated database develops empirical relationships among various source parameters and estimates earthquake magnitude using machine learning techniques. In the past, empirical relations between various parameters were thought to depend on the tectonic setting (Haiyun & Xiabin, 2003; Wells & Coppersmith, 1994). However, later it was found that separating the data by tectonic setting produces no statistically different results and does not improve the statistical significance of the regression (Wells & Coppersmith, 1994), whereas Ghassemi (2016) concluded that earthquake source-scaling properties depend on the faulting style. Hence, based on faulting regimes, we classified the data into three broad categories: strike-slip, reverse or normal earthquakes. The oblique-slip events are categorised as one of the three faulting types and are not explicitly examined.

Earthquakes with poor data quality or providing inadequate information were assessed but not included in this study. A few earthquakes in the database have incomplete fault parameters and moment magnitude information. The solution to this problem has been discussed in detail in the subsequent section.

3. Methodology

In this study, we first use the conventional regression-based methodology to find meaningful relation between the various fault parameters and earthquake magnitudes. Then, using the machine learning based ANN and GBM approaches, we compare the results obtained by both methods, i.e.

machine-learning methods and the conventional methodology, and finally check the results regarding consistency, accuracy and robustness in predicting the earthquake magnitude.

3.1. Regression Analysis

There are numerous regression methods available to investigate the relationship between two variables. In this study, we use ordinary least square regression analysis to derive log-linear relationships between fault parameters and magnitudes of earthquakes (Wells & Coppersmith, 1994). These relationships, however, are derived separately for each fault parameter; independent regression equations relating to earthquake magnitude are obtained. We perform our regressions for the updated catalogue described in the data section. The number of events, regression coefficients (a and b), standard error of the coefficients, standard deviation of the dependent variable (s) and correlation coefficient are all regression descriptors (r). The empirical relationships are of the form $y = a + b \cdot \log(x)$ or $\log(y) = a + b \cdot \log(x)$, with y being the dependent variable and x being the independent variable. The application will determine the independent and dependent variables, either the expected magnitude for a given fault parameter or the expected fault parameter for a given magnitude. The strengths and limitations of using the ordinary least square regression are described elsewhere (Wells & Coppersmith, 1994). We also plotted the regression relationship derived earlier (Wells & Coppersmith, 1994) to check the validity of the revised relationships, which include the events associated with subduction zones, oceanic and continental plate boundaries, and oceanic plates.

3.2. Artificial Neural Network (ANN) Regression

Artificial neural networks (ANN) are inspired by biological neural networks, which comprise neurons and weighted connections between distinct layers of neurons (Hassoun, 1995). Theoretically, ANN is a computational model that may establish linear or

nonlinear relationships between input and output dataset variables (Nedic et al., 2014). ANN have many advantages over statistical approaches (Zhang et al., 1998). Therefore, ANN can achieve high accuracy. However, because compiling a large dataset can be difficult for the user, improving accuracy is always challenging (Malakar et al., 2022).

Neural networks use artificial synapses to connect neurons that store and process data. The primary function of neurons is to accept information, process it computationally, output information and form neurons by connecting other neurons. The training set is used by the learning algorithm to create suitable network weights that connect the input and output layers. A neural network can be trained using a variety of algorithms, which vary depending on the type and topology of the neural network. The feed-forward networks are the most well-known ANN topology. Multilayer perceptron (MLP), radial basis function neural networks (RBFNN) and general regression neural networks (GRNN) are several types of feed-forward neural networks. This paper uses MLP modelling to predict earthquake magnitude based on fault parameters.

3.3. Gradient Boosting Machine (GBM) Regression

Friedman (2001) proposed the gradient boosting machine (GBM) as an ensemble method for regression and classification problems. The technique's primary heuristic is to build new base learners that are maximally correlated with the negative gradient of the loss function related to the entire ensemble. It iterates to improve the system performance on weak points. It generated trees sequentially to correct the preceding errors. GBM can help to reduce bias but with a high variance. It is a sequential process with challenges to train the information, but it outperforms random forest (Olson et al., 2016).

The GBM begins with a single leaf at first. This single leaf represents the preliminary guess for the predicted value. Sequentially, the GBM creates the complete tree. In GBM, a tree is determined by the errors made by the preceding tree of a fixed size. GBM handles the tree scaling, and the same amount

scales all trees. This process continues until the user-specified/number of trees or other trees fail to improve it. GBM solves this problem by using a learning rate to scale the contribution of the new tree. The value of the learning rate ranges from 0 to 1. Scaling the tree using the learning rate improves the performance of the GBM. The algorithm of GBM is described in detail by Friedman (2001). The number of trees, learning rate and variable interactions influence the GBM model's performance. One can achieve optimal model performance by selecting the best combination of these parameters. The GBM regression model implemented in this study is provided by Wagner et al. (2014).

3.4. Application of ANN and GBM Regression

As discussed above, we use two machine-learning techniques, i.e., ANN and GBM. The main inputs to the proposed models are the fault parameters. The output of the models is earthquake magnitude. The trained model optimally maps a given set of fault parameters to the actual earthquake magnitude and predicts the magnitude of an earthquake for unseen data; 70% of the data points were used for training and 30% for testing. In the training process, the training data points were used, and the model was adjusted based on their error. Remember that the main idea is not the model accuracy but the prediction error. In practical applications, only RLD-RW-RA might be available as an input, whereas the maximum slip value is often unknown (Asim et al., 2019). However, we calculated the value of the unknown/unreliable fault parameters of the particular events using the revised relationships developed for all slip types and utilised them as inputs for the machine-learning models; after checking the reliability of the estimated parameters for earthquakes that have incomplete moment magnitude information, surface wave magnitude (M_s) has been used. M_s and M have no consistent difference up to magnitude 8.9 in our data set (Fig. 2). The correlation coefficient between each pair of M_s and M values is 0.97, while the standard deviation is 0.23.

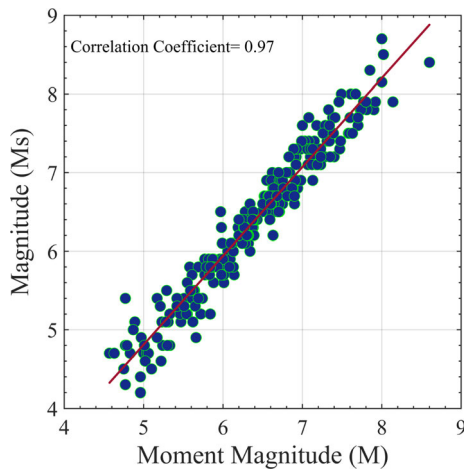


Figure 2

The linear regression between M_s (surface wave magnitude), and the moment magnitude (M) for global historical events

The activation function used in the neural network for the output layer differs according to the problem under consideration. This function is typically chosen to be linear for regression problems (Thomas & Suhner, 2015). The number of nodes in the hidden layer is another critical step in developing the MLP model. This problem is highly controversial, and there is no rule of thumb. Linoff and Berry (2011) concluded that the number of nodes in the hidden

layer should be between the number of nodes in the input layer and twice that number. Since there are six inputs, the number of hidden nodes in our case is between 7 and 12. The model parameters for the GBM regression are learning rate = 0.02 maximum tree size = 10 and loss function = squared error loss.

Finally, the performance of the models was estimated. A cross-validation strategy is applied to test the predictions of these techniques. The k-fold cross-validation method is commonly used to demonstrate the performance of regression techniques (Wong, 2015). Therefore, we have used this approach with $k = 10$, tenfold cross-validation (Idris et al., 2017). The result of the analysis is discussed in the following sections.

4. Results and Discussions

4.1. Regression Result

In this study, we established empirical relationships among various earthquake source parameters, i.e. M, SRL, RLD, RW, RA, AD and MD, using an extensive and updated database of global earthquakes (Figs. 3, 4, 5, 6, 7, 8, 9 and 10). Additionally, empirical relationships and correlation coefficients

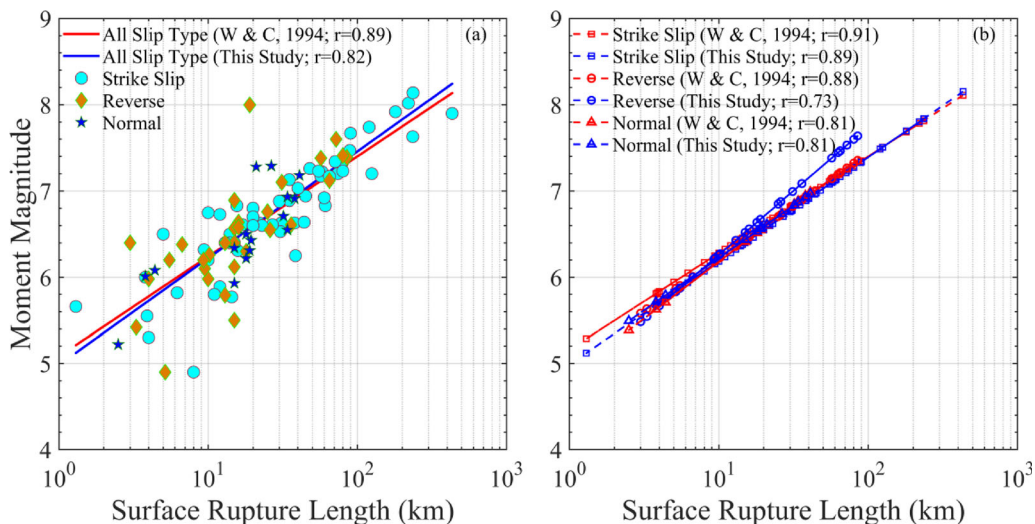


Figure 3

Regression analysis between surface rupture length and the moment magnitude: **a)** for all slip types, and **b)** for the strike-slip, reverse and normal-slip earthquakes

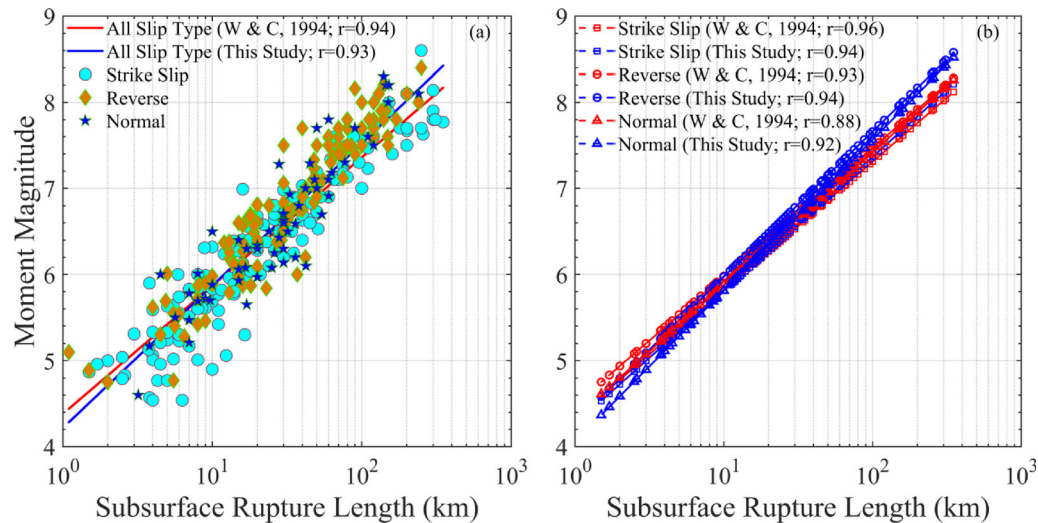


Figure 4

Regression analysis between subsurface rupture length and the moment magnitude: **a)** for all slip types, and **b)** for the strike-slip, reverse and normal-slip earthquakes

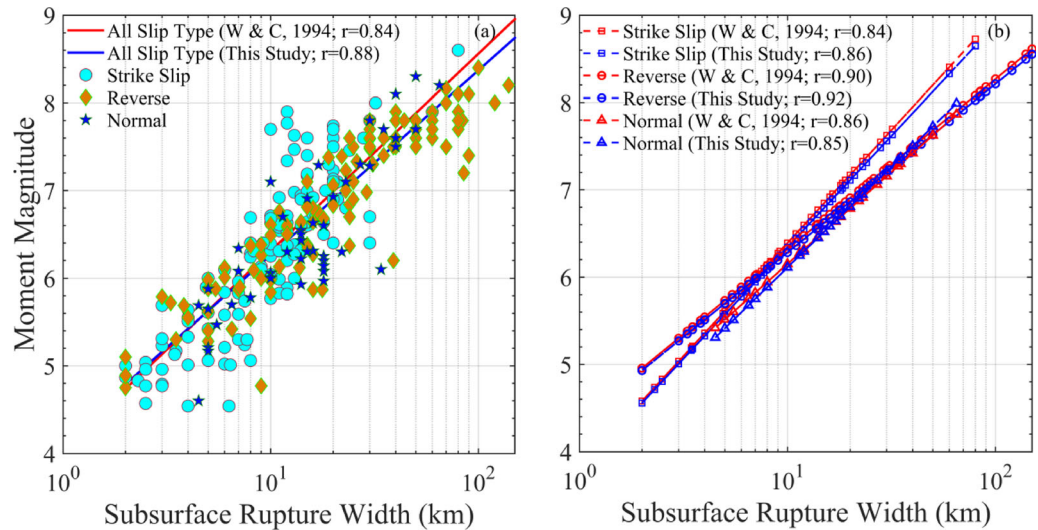


Figure 5

Regression analysis between down dip rupture width and the moment magnitude: **a)** for all-slip types, and **b)** for the strike-slip, reverse and normal-slip earthquakes

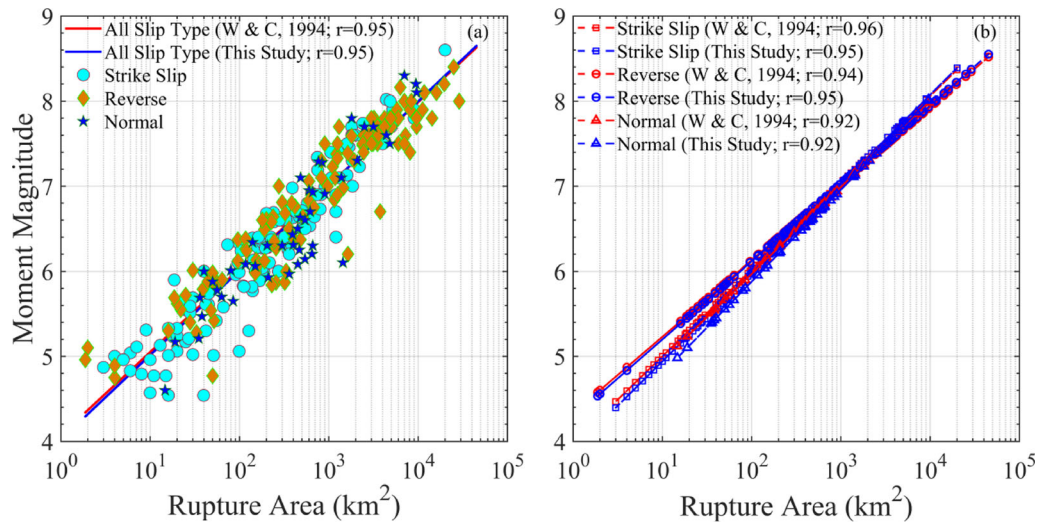


Figure 6

Regression analysis between rupture area and the moment magnitude: **a)** for all-slip types, and **b)** for the strike-slip, reverse and normal-slip earthquakes

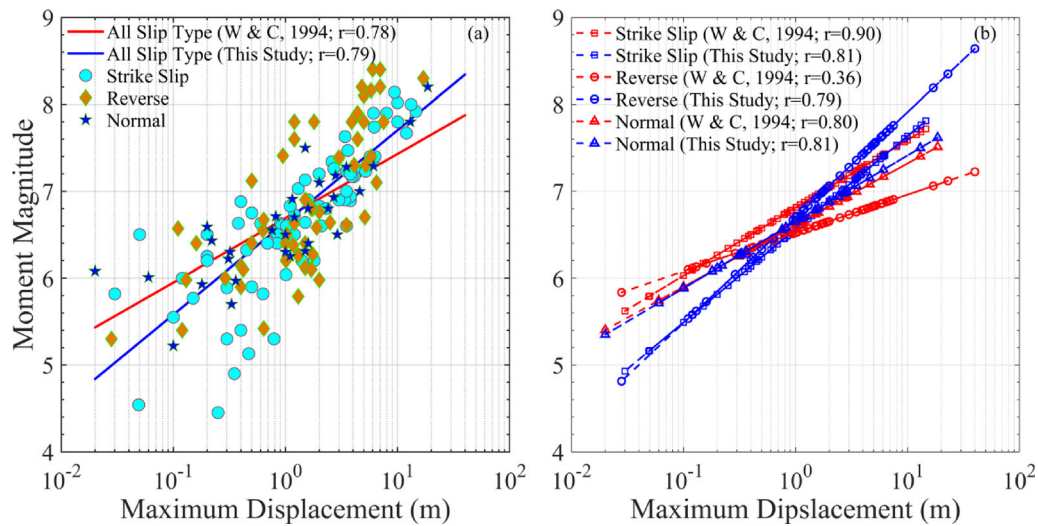


Figure 7

Regression analysis between maximum displacement and the moment magnitude: **a)** for all-slip types, and **b)** for the strike-slip, reverse and normal-slip earthquakes

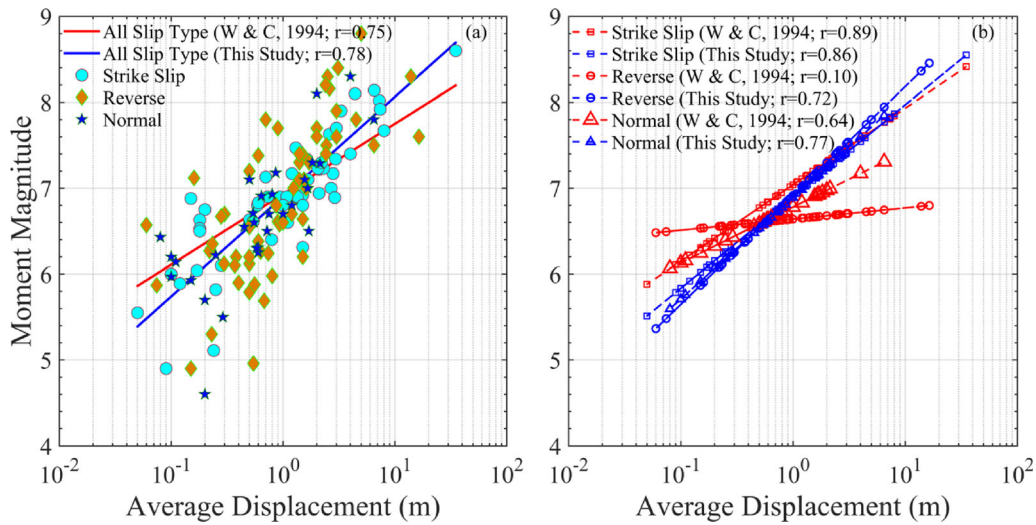


Figure 8

Regression analysis between average displacement and the moment magnitude: **a)** for all slip types, and **b)** for the strike-slip, reverse and normal-slip earthquakes

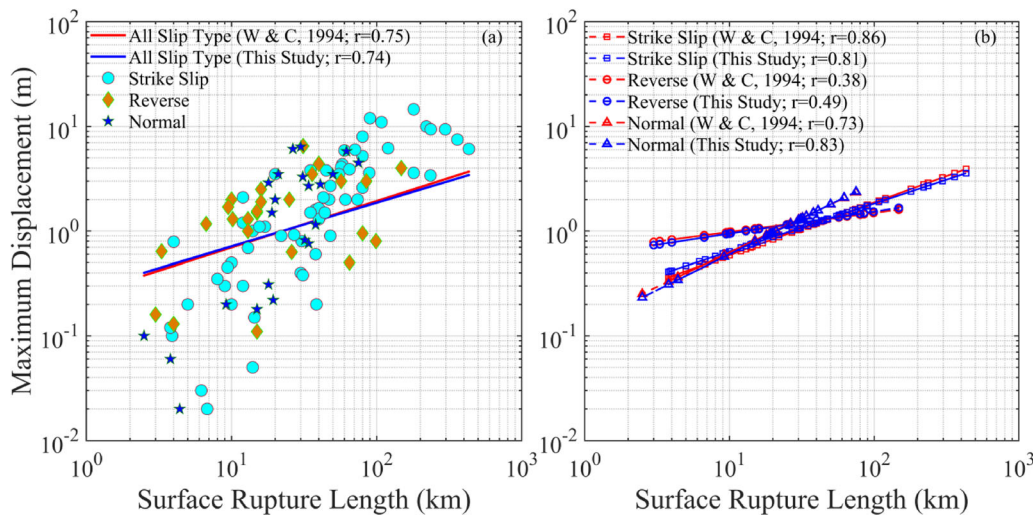


Figure 9

Regression analysis between surface rupture length and the maximum displacement: **a)** for all slip types, and **b)** for the strike-slip, reverse and normal-slip earthquakes

are determined independently for all fault types, strike-slip, reverse and normal faults. Finally, the

results were compared with those obtained in the previous studies.

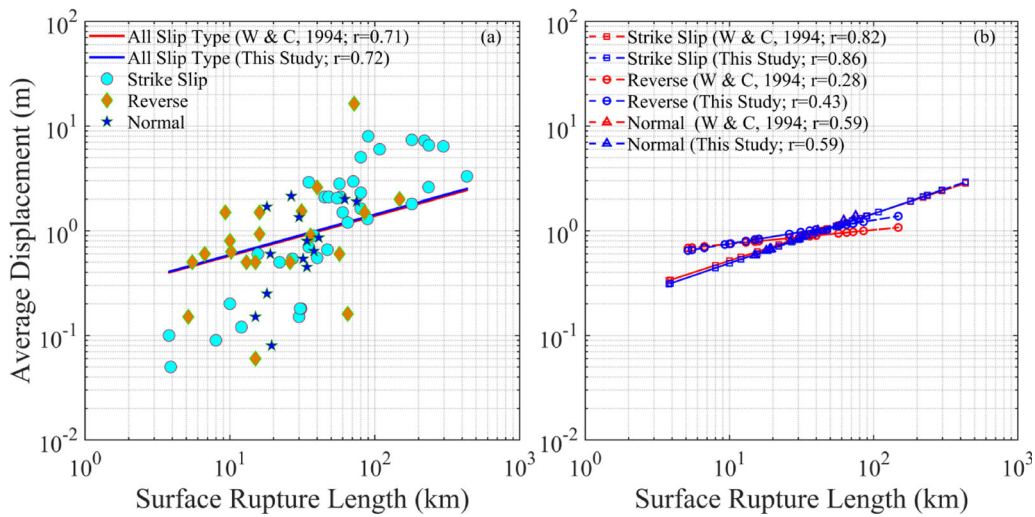


Figure 10

Regression analysis between surface rupture length and the average displacement: **a)** for all slip types, and **b)** for the strike-slip, reverse and normal-slip earthquakes

Our analysis shows that correlation coefficients (r) for the aforementioned parameters of the revised catalogue range from 0.43 to 0.95 (Tables 1 and 2). The small size of the datasets for a particular earthquake fault may contribute to the poor correlation. Wells and Coppersmith (1994) suggested that datasets with 10 to 14 data points may not show significant correlations. However, it is found that results are consistent with those of Wells and Coppersmith (1994), except for the reverse faulting of maximum and average displacement (Tables 1 and 2). Furthermore, Wells and Coppersmith (1994) have fewer data points for the reverse faulting of maximum and average displacement that may not show significant correlations. Adding events associated with subduction zones, oceanic and continental plate boundaries, and oceanic plates in the catalogue has significantly improved the correlation coefficient, especially for the reverse fault.

Correlation coefficients between moment magnitude and fault parameters for various slip types are different (Tables 1, 2). Furthermore, Tables 1 and 2 indicates that all correlation coefficients for the subsurface rupture length and rupture area are more than 0.90. The strike-slip and reverse faults are highly correlated in both the cases. Compared to Wells and

Coppersmith (1994), trivial improvements in the correlation coefficient were noticed, especially for the rupture area. For rupture width, the correlation coefficients are more than or equal to 0.80, with a maximum value of 0.92 for reverse fault. Furthermore, an improved correlation coefficient was noticed except for normal faulting, which shows a minor decreasing result. For surface rupture length, we found that the correlation coefficient ranges from 0.73 to 0.89 and decreases compared with previous literature, especially for the reverse fault earthquakes. For maximum and average slip, the correlation coefficient ranges from 0.79 to 0.81 and 0.72 to 0.86, respectively. In both cases, the correlation coefficient is increased and revised, and a significant improvement can be seen for reverse fault, except for strike-slip, which shows a decreasing result. Generally, the relatively high correlations ($r > 0.7$) and low standard deviations obtained by the regression analysis imply a strong correlation. Hence, these revised equations derived in this study may be used to calculate dependent variables in case of various types of earthquakes.

For the graphical representation of the result, each pair of parameters are represented by two plots. The first plot illustrates the all-slip-type regression line

Table 1

Regression relations between various source parameters and moment magnitude (M) for worldwide earthquakes

Relations	Slip types	Number of events	Coefficients		Standard errors		Standard Deviation	Correlation coefficient (r)	Correlation coefficient (WC94)
			a	b	sa	sb			
$M = a + b \cdot \log(SRL)$	All	106	4.98	1.24	0.12	0.08	0.41	0.82	0.89
	SS	59	4.98	1.20	0.12	0.08	0.32	0.89	0.91
	R	31	4.77	1.48	0.33	0.26	0.57	0.73	0.88
	N	16	5.01	1.21	0.30	0.24	0.34	0.81	0.81
$\log(SRL) = a + b \cdot M$	All	106	-2.27	0.54	0.25	0.04	0.27	0.82	0.89
	SS	59	-2.99	0.66	0.30	0.04	0.24	0.89	0.91
	R	31	-1.15	0.36	0.41	0.06	0.28	0.73	0.88
	N	16	-2.26	0.54	0.68	0.11	0.23	0.81	0.81
$M = a + b \cdot \log(RLD)$	All	334	4.22	1.65	0.05	0.03	0.33	0.93	0.94
	SS	152	4.26	1.55	0.07	0.05	0.32	0.94	0.96
	R	124	4.29	1.69	0.09	0.05	0.30	0.94	0.93
	N	58	4.06	1.75	0.15	0.10	0.33	0.92	0.88
$\log(RLD) = a + b \cdot M$	All	334	-2.04	0.53	0.07	0.01	0.19	0.93	0.94
	SS	152	-2.26	0.57	0.11	0.02	0.19	0.94	0.96
	R	124	-2.08	0.53	0.12	0.02	0.17	0.94	0.93
	N	58	-1.72	0.48	0.18	0.03	0.17	0.92	0.88
$M = a + b \cdot \log(RW)$	All	294	4.16	2.11	0.08	0.07	0.44	0.88	0.84
	SS	134	3.79	2.55	0.14	0.13	0.47	0.86	0.84
	R	113	4.35	1.93	0.11	0.08	0.36	0.92	0.90
	N	47	3.80	2.31	0.25	0.21	0.44	0.85	0.86
$\log(RW) = a + b \cdot M$	All	294	-1.29	0.37	0.08	0.01	0.19	0.88	0.84
	SS	134	-0.83	0.29	0.10	0.02	0.16	0.86	0.84
	R	113	-1.71	0.44	0.12	0.02	0.17	0.92	0.90
	N	47	-0.88	0.32	0.19	0.03	0.16	0.85	0.86
$M = a + b \cdot \log(RA)$	All	295	4.02	1.00	0.05	0.02	0.31	0.95	0.95
	SS	133	3.90	1.04	0.07	0.03	0.29	0.95	0.96
	R	116	4.28	0.92	0.09	0.03	0.30	0.95	0.94
	N	46	3.72	1.08	0.19	0.07	0.34	0.92	0.92
$\log(RA) = a + b \cdot M$	All	295	-3.34	0.90	0.12	0.02	0.29	0.95	0.95
	SS	133	-3.13	0.86	0.16	0.02	0.26	0.95	0.96
	R	116	-3.89	0.98	0.22	0.03	0.31	0.95	0.94
	N	46	-2.51	0.78	0.33	0.05	0.29	0.92	0.92
$M = a + b \cdot \log(MD)$	All	163	6.64	1.06	0.04	0.06	0.50	0.79	0.78
	SS	73	6.56	1.07	0.06	0.09	0.48	0.81	0.90
	R	59	6.70	1.21	0.08	0.13	0.56	0.79	0.36
	N	31	6.65	0.76	0.07	0.10	0.37	0.81	0.80
$\log(MD) = a + b \cdot M$	All	163	-3.89	0.59	0.24	0.04	0.38	0.79	0.78
	SS	73	-3.99	0.61	0.35	0.05	0.36	0.81	0.90
	R	59	-3.34	0.51	0.37	0.05	0.36	0.79	0.36
	N	31	-5.71	0.86	0.77	0.12	0.39	0.81	0.80
$M = a + b \cdot \log(AD)$	All	144	6.90	1.16	0.04	0.08	0.50	0.78	0.75
	SS	53	6.90	1.07	0.05	0.09	0.37	0.86	0.89
	R	59	6.91	1.27	0.08	0.16	0.61	0.72	0.10
	N	32	6.89	1.18	0.09	0.18	0.47	0.77	0.64
$\log(AD) = a + b \cdot M$	All	144	-3.62	0.52	0.24	0.04	0.33	0.78	0.75
	SS	53	-4.82	0.70	0.40	0.06	0.30	0.86	0.89
	R	59	-2.81	0.40	0.36	0.05	0.34	0.72	0.10
	N	32	-3.56	0.50	0.51	0.08	0.31	0.77	0.64

(*M*: moment magnitude, *SRL*: surface rupture length (km), *RLD*: subsurface rupture length (km), *RW*: downdip rupture width (km), *RA*: rupture area (km²), *MD*: maximum displacement (m), *AD*: average displacement (m), *SS*: strike slip, *R*: reverse, and *N*: normal faults)

Table 2

Regression relations between surface rupture length (SRL) and displacement (MD) for worldwide earthquakes

Relations	Slip types	Number of events	Coefficients		Standard errors		Standard Deviation	Correlation coefficient (r)	Correlation coefficient (WC94)
			a	b	sa	sb			
$\log (MD) = a + b \cdot \log (SRL)$	All	109	-1.29	0.96	0.13	0.08	0.43	0.74	0.75
	SS	62	-1.50	1.05	0.16	0.10	0.39	0.81	0.86
	R	25	-0.54	0.49	0.25	0.18	0.41	0.49	0.38
	N	22	-2.08	1.57	0.33	0.24	0.42	0.83	0.73
$\log (SRL) = a + b \cdot \log (MD)$	All	109	1.39	0.57	0.03	0.05	0.33	0.74	0.75
	SS	62	1.47	0.62	0.04	0.06	0.30	0.81	0.86
	R	25	1.26	0.49	0.08	0.18	0.41	0.49	0.38
	N	22	1.32	0.43	0.05	0.07	0.22	0.83	0.73
$\log (AD) = a + b \cdot \log (SRL)$	All	71	-1.42	0.89	0.17	0.10	0.39	0.72	0.71
	SS	37	-1.81	1.09	0.19	0.11	0.32	0.86	0.82
	R	20	-0.81	0.52	0.36	0.26	0.47	0.43	0.28
	N	14	-1.99	1.24	0.72	0.49	0.37	0.59	0.59
$\log (SRL) = a + b \cdot \log (AD)$	All	71	1.58	0.58	0.04	0.07	0.32	0.72	0.71
	SS	37	1.67	0.68	0.04	0.07	0.25	0.86	0.82
	R	20	1.38	0.36	0.09	0.18	0.39	0.43	0.28
	N	14	1.52	0.28	0.05	0.11	0.17	0.59	0.59

SRL: surface rupture length; *MD*: Max. displacement; *AD*: average displacement; Wells & Coppersmith, 1994

(all data regression fit) (Figs. 3a, 4, 5, 6, 7, 8, 9, and 10a). The second plot depicts the regression lines for individual slip types (Figs. 3b, 4, 5, 6, 7, 8, 9, and 10b). The regression line length represents the data range for each empirical relationship. Comparisons with previous studies (Wells & Coppersmith, 1994) for all slip types, surface rupture length, subsurface rupture length, rupture width and rupture area show a minor difference (Figs. 3a, 4, 5, and 6a). The relationships between maximum and average displacement with a moment magnitude show considerable differences between the positions of all slip type regression lines (Figs. 7a, 8a). Our analysis also noticed that surface rupture length, subsurface rupture length and rupture area differ little in the position of the regression lines as a function of the sense of slip (Figs. 3b, 4b, and 6b). However, it is recommended to use regressions specified for the faulting style instead of that derived using data for all slip types (Ghassemi, 2016; Stirling et al., 2013).

These empirical relationships can be used to calculate the maximum possible earthquake magnitudes for a given fault zone. If the input fault parameter is a maximum value, then the assumption

that a given earthquake magnitude is a maximum value holds. However, because dispersion is associated with statistical relations measured by standard deviation, any event with the given rupture parameters can have higher and lower magnitudes.

The regressions for subsurface rupture length and rupture area can estimate earthquake magnitudes on subsurface seismic sources, which cannot be assessed from surface data (Wells & Coppersmith, 1994). Furthermore, the rupture width relationship estimates additional magnitude if the fault zone's downdip geometry is known. Detailed geological studies along a fault zone can estimate the average and maximum displacement of particular paleoseismic events. Regressions can estimate magnitudes from these displacement estimates. These considerations and assessments may be combined into a probabilistic distribution of the maximum magnitude for seismic hazard analyses. Additionally, these regressions have potential engineering applications, particularly in seismic design criteria for the facility, where estimates of the displacement that may occur when the facility crosses a fault are required.

Table 3
Performance of ANN, GBM and conventional regression methods

Performance measures	ANN	GBM	Regression method (all slip type)					
	SRL-RLD-RW-RA-MD-AD	SRL-RLD-RW-RA-MD-AD	SRL (this study)	RLD (this study)	RW (this study)	RA (this study)	MD (this study)	AD (this study)
MAE	0.136	0.169	0.276	0.215	0.279	0.194	0.341	0.345
RMSE	0.182	0.230	0.354	0.294	0.392	0.272	0.435	0.434

4.2. ANN and GBM Regression

In this section, we present results of the ANN and GBM methods to illustrate that the machine learning technique can potentially improve seismic hazard assessments, using magnitude predictions based on fault parameters, compared with those obtained by the conventional methodologies. The magnitude estimation performance is expressed in terms of root means square error (RMSE) and mean absolute error (MAE). These errors are calculated between actual and estimated magnitudes to quantify the overall performance of the ANN, GBM and conventional regression techniques.

The regression relations can be used to estimate earthquake magnitude based on a single fault parameter, which could be extremely useful for paleoseismological studies. The lack of self-consistency is a significant issue in such approaches (Asim et al., 2019). However, the ANN- and GBM-based earthquake magnitude estimation algorithms use all fault parameters simultaneously. Considering potential nonlinear parameter correlations, the simultaneous use of all fault parameters eliminates the need for different magnitude estimations, and a self-consistent magnitude estimation is achieved. The inconsistency and errors of regression-based estimates for SRL, RLD, RW, RA, MD and AD equations are shown in Table 3. Contrarily, the ANN and GBM simultaneously consider all given fault parameters and provide a single prediction. The RMSE value of the ANN is 0.182; for GBM, the value is about 0.230. The regression based on SRL, RLD, RW, RA, MD and AD indicates a value of 0.354, 0.294, 0.392, 0.272, 0.435 and 0.434, respectively. The results show that GBM outperforms regression equations, but ANN outperforms both methodologies in terms of

accuracy. Machine learning predictions reveal that ANN and GBM perform well across the data set. Figure 11 displays scatter plots of the actual magnitudes and regression, GBM and ANN-based predictions. The figure reveals that the simultaneous use of all fault parameters significantly decreases errors compared to the regression equations. We also detect systematic deviations from the actual magnitude and the regression-based approach for surface rupture length, maximum and average displacement (Figs. 11a, e, f). The reason may be less that the dataset has been utilised for developing the regression among moment magnitude and maximum and average displacement (Table 1). We also observe that the RA performs better than other fault parameters, followed by RLD for the potential earthquake magnitude prediction as a function of a single fault parameter (Table 3). Hence, using the relationship between moment magnitude and rupture area is better suited to estimate the potential magnitude in the case of conventional regression.

ANN and GBM model training requires high-quality, quantity and diverse data. More extensive and diversified data sets improve model training with robust prediction capabilities (Asim et al., 2019). The dataset used in this study contains earthquakes with magnitudes ≥ 4.5 . However, the catalogue is skewed toward higher magnitude events and includes fewer low magnitude events. The reason may be that lower-magnitude earthquakes were not frequently reported during the pre-instrumental era. Our catalogue has varying numbers of samples in different magnitude classes. We define various magnitude ranges for predicted earthquakes and examine prediction performance based on magnitude bin sample size. The result is presented in terms of MAE, and RMSE has

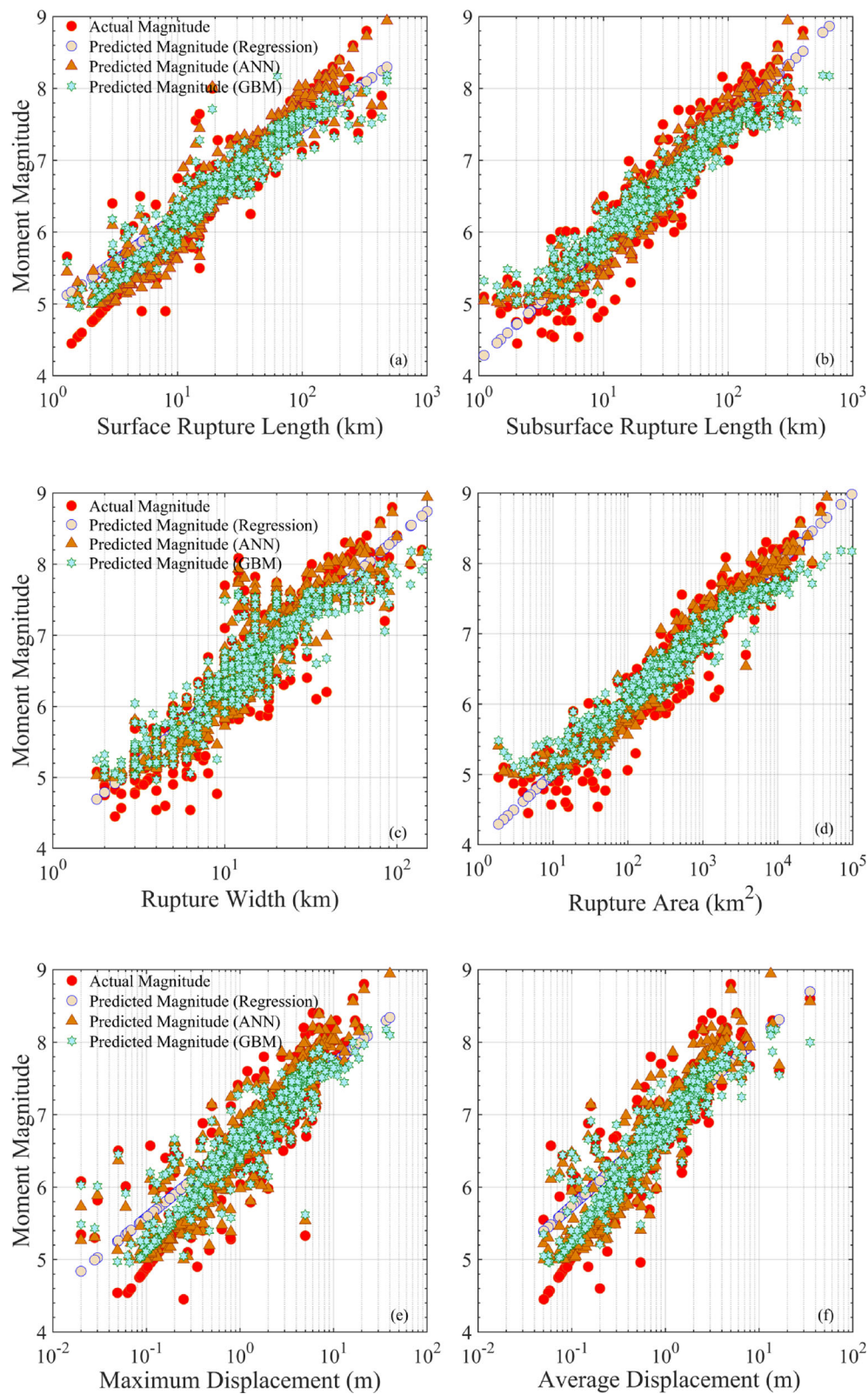


Figure 11

Actual and predicted earthquake magnitudes using regression, ANN and GBM plotted against **a)** surface rupture length, **b)** subsurface rupture length, **c)** width, **d)** rupture area, **e)** maximum displacement and **f)** average displacement

been tabulated in Table 4. The MAE and RMSE are the highest for the least abundant predicted magnitude class, which can be noted for the prediction by the GBM for magnitude > 8. The magnitude prediction by ANN is relatively consistent and more robust, especially for higher-magnitude earthquakes. In contrast, the prediction by GBM is less consistent based on the sample size in the magnitude bins. The GBM shows a robust result for the magnitude bin of 6.0–7.0 and is significantly less accurate for great earthquakes than ANN (Table 4). The reason may be due to the different algorithms used by these machine learning techniques. Asim et al. (2019) also compared the performance of ANN with another machine-learning technique, i.e., the random forest method. Their result suggests that ANN provides more consistent and robust results than the random forest model. Hence, it is recommended to use the ANN machine-learning technique for estimating the earthquake magnitude using all available fault parameters simultaneously for consistent and robust results. Data diversity is one of the ways to improve the machine-learning models' accuracy.

5. Conclusions

In this study, we have used extended earthquake catalogue than those used in previous studies, by

adding recent earthquakes source parameters data from published literatures. Using regression analysis and machine learning approach, we have also derived new empirical relations between various source and rupture parameters. We apply log-linear regression to account for measurement errors across all variables and assume to be established mutually self-consistent correlations. Our analysis shows that the correlation coefficients (r) between various parameters for the revised catalogue used in this study vary between 0.43 and 0.95, which were between 0.10 and 0.96 in earlier studies (Wells & Coppersmith, 1994). The results also indicate that adding events associated with subduction zones, oceanic and continental plate boundaries, and earthquakes from oceanic plates in the catalogue can significantly improved the correlation coefficient, especially for the reverse fault earthquakes.

The log-linear regression may not account for the nonlinear behaviour of rupture parameters. These equations are independently used for each fault parameter, leading to inconsistency in magnitude prediction. Hence, we have applied machine learning models, i.e., artificial neural network and gradient boosting machine regression, for fault-based earthquake magnitude estimation. Machine learning algorithms can identify hidden patterns in data and use all fault parameters simultaneously to ensure consistency. The results show that GBM estimates earthquake magnitude better than regression equations, but ANN outperforms both. These models can also incorporate additional information, encouraging researchers to use additional fault parameters in the future. The result of this study may be helpful for engineering applications, geological studies along

Table 4
Prediction quality for different magnitude ranges of ANN and GBM

	ANN				GBM			
Performance measures	[4.5, 6.0)	[6.0, 7.0)	[7.0, 8.0)	≥ 8.0	[4.5, 6.0)	[6.0, 7.0)	[7.0, 8.0)	≥ 8.0
Number of instances in catalogue	122	185	141	28	122	185	141	28
Number of instances predicted	131	168	155	22	108	216	148	4
MAE	0.180	0.130	0.110	0.105	0.213	0.108	0.206	0.887
RMSE	0.235	0.181	0.133	0.125	0.260	0.151	0.255	0.903

fault zones, and understanding paleoseismic earthquakes where it is often not possible to obtain reliable magnitude estimates. Furthermore, results could also be applied to complement existing efforts for real-time estimation of earthquake characteristics, which are significant for improving early tsunami warning systems.

Acknowledgements

The authors thank the Indian Institute of Technology Kharagpur and the Ministry of Education, Government of India, for providing all necessary support. We thank various agencies and researchers who have made their findings publicly available in the form of research articles that may have been used in this study. We gratefully acknowledge the handling editor and the anonymous reviewers for their insightful comments that significantly improved the original manuscript.

Author Contributions SM and AKR conceived the idea. SM collected and analysed data, and AKR and AKG supervised the project. VKK helped in revising the manuscript. All authors discussed the results and contributed to the final manuscript.

Data Availability

All data used in this study are publicly available.

Declarations

Conflict of interest The authors declare no competing interests.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

REFERENCES

- Acharya, H. K. (1979). Regional variations in the rupture-length magnitude relationships and their dynamical significance. *Bulletin of the Seismological Society of America*, 69, 2063–2084.
- Asim, K. M., Idris, A., Iqbal, T., & Martínez-Álvarez, F. (2018). Earthquake prediction model using support vector regressor and hybrid neural networks. *PLoS One*, 13, e0199004. <https://doi.org/10.1371/journal.pone.0199004>
- Asim, K. M., Javed, F., Hainzl, S., & Iqbal, T. (2019). Fault parameters-based earthquake magnitude estimation using artificial neural networks. *Seismological Research Letters*, 90, 1544–1551. <https://doi.org/10.1785/0220190051>
- Atkinson, G. M., & Boore, D. M. (2003). Empirical ground-motion relations for subduction-zone earthquakes and their application to cascadia and other regions. *Bulletin of the Seismological Society of America*, 93, 1703–1729. <https://doi.org/10.1785/0120020156>
- Atzori, S., Antonioli, A., Tolomei, C., De Novellis, V., De Luca, C., & Monterroso, F. (2019). InSAR full-resolution analysis of the 2017–2018 M>6 earthquakes in Mexico. *Remote Sensing of Environment*. <https://doi.org/10.1016/j.rse.2019.111461>
- Barbot, S., Luo, H., Wang, T., Hamiel, Y., Piatibratova, O., Javed, M. T., Braitenberg, C., & Gurbuz, G. (2023). Slip distribution of the February 6, 2023 Mw 7.8 and Mw 7.6, Kahramanmaraş, Turkey earthquake sequence in the East Anatolian Fault Zone. *Seismica*. <https://doi.org/10.26443/seismica.v2i3.502>
- Blaser, L., Krüger, F., Ohrnberger, M., & Scherbaum, F. (2010). Scaling relations of earthquake source parameter estimates with special focus on subduction environment. *Bulletin of the Seismological Society of America*, 100, 2914–2926. <https://doi.org/10.1785/0120100111>
- Boncio, P., Liberi, F., Caldarella, M., & Nurminen, F.-C. (2018). Width of surface rupture zone for thrust earthquakes: Implications for earthquake fault zoning. *Natural Hazards and Earth System Sciences*, 18, 241–256. <https://doi.org/10.5194/nhess-18-241-2018>
- Bonilla, M. G., Mark, R. K., & Lienkaemper, J. J. (1984). Statistical relations among earthquake magnitude, surface rupture length, and surface fault displacement. *USGS Open-File Report*, 84–256.
- Chanda, S., & Somala, S. N. (2021). Single-component/single-station-based machine learning for estimating magnitude and location of an earthquake: A support vector machine approach. *Pure and Applied Geophysics*, 178, 1959–1976. <https://doi.org/10.1007/s00024-021-02745-8>
- CIESIN (2018). Gridded Population of the World, Version 4 (GPWv4): Population Density Adjusted to Match 2015 Revision UN WPP Country. *NASA Socioeconomic Data and Applications Center (SEDAC)*. <https://doi.org/10.7927/H4F47M65>. Accessed 21st June 2023.
- Das, R., & Meneses, C. (2021). A unified moment magnitude earthquake catalog for Northeast India. *Geomatics, Natural Hazards and Risk*, 12, 167–180. <https://doi.org/10.1080/19475705.2020.1863269>
- Dhanya, J., & Raghukanth, S. T. G. (2018). Ground motion prediction model using artificial neural network. *Pure and Applied Geophysics*, 175, 1035–1064. <https://doi.org/10.1007/s00024-017-1751-3>
- Fan, W., Wei, S. S., Tian, D., McGuire, J. J., & Wiens, D. A. (2019). Complex and diverse rupture processes of the 2018 Mw

- 8.2 and Mw 7.9 Tonga-Fiji deep earthquakes. *Geophysical Research Letters*, 46, 2434–2448. <https://doi.org/10.1029/2018GL080997>
- Friedman, J. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*. <https://doi.org/10.1214/aos/1013203451>
- Fujii, Y., & Matsu'ura, M. (2000). Regional difference in scaling laws for large earthquakes and its tectonic implication. *Pure and Applied Geophysics*, 157, 2283–2301. <https://doi.org/10.1007/PL00001085>
- Ghassemi, M. R. (2016). Surface ruptures of the Iranian earthquakes 1900–2014: Insights for earthquake fault rupture hazards and empirical relationships. *Earth-Science Reviews*, 156, 1–13. <https://doi.org/10.1016/j.earscirev.2016.03.001>
- Gomez, B., & Kadri, U. (2021). Earthquake source characterization by machine learning algorithms applied to acoustic signals. *Scientific Reports*, 11, 23062. <https://doi.org/10.1038/s41598-021-02483-w>
- Gomez, B., & Kadri, U. (2023). Numerical validation of an effective slender fault source solution for past tsunami scenarios. *Physics of Fluids*. <https://doi.org/10.1063/5.0144360>
- Guo, R., Yang, H., Zhu, Y., Zheng, Y., Xu, J., Zhang, L., & An, C. (2021). Narrow rupture of the 2020 Mw 7.4 La Crucecita, Mexico. *Earthquake. Seismological Research Letters*, 92, 1891–1899. <https://doi.org/10.1785/0220200328>
- Haiyun, W., & Xiaxin, T. (2003). Relationships between moment magnitude and fault parameters: Theoretical and semi-empirical relationships. *Earthquake Engineering and Engineering Vibration*, 2, 201–211. <https://doi.org/10.1007/s11803-003-0004-x>
- Hassoun, M. H. (1995). *Fundamentals of Artificial Neural Networks*. The MIT Press Cambridge Massachusetts.
- Hayes, G. P., Myers, E. K., Dewey, J. W., Briggs, R. W., Earle, P. S., Benz, H. M., Smoczyk, G. M., Flamme, H. E., Barnhart, W. D., Gold, R. D., & Furlong, K. P. (2017). Tectonic summaries of magnitude 7 and greater earthquakes from 2000 to 2015. *U.S. Geological Survey Open-File Report 2016–1192*. <https://doi.org/10.3133/ofr20161192>
- Idris, A., Iftikhar, A., & Rehman, Z. U. (2017). Intelligent churn prediction for telecom using GP-AdaBoost learning and PSO undersampling. *Cluster Computing*, 22, 7241–7255. <https://doi.org/10.1007/s10586-017-1154-3>
- Jiang, Y., González, P. J., & Bürgmann, R. (2022). Subduction earthquakes controlled by incoming plate geometry: The 2020 M > 7.5 Shumagin, Alaska, earthquake doublet. *Earth and Planetary Science Letters*. <https://doi.org/10.1016/j.epsl.2022.117447>
- Kim, B.-H., Rehman, K., Cho, Y.-K., & Hong, S. H. (2023). Tsunami waveform forecasting at cooling water intakes of nuclear reactors with deep learning model. *Physics of Fluids*. <https://doi.org/10.1063/5.0156882>
- Konstantinou, K., Papadopoulos, G., Fokaefs, A., & Orfanogianaki, K. (2005). Empirical relationships between aftershock area dimensions and magnitude for earthquakes in the Mediterranean Sea region. *Tectonophysics*, 403, 95–115. <https://doi.org/10.1016/j.tecto.2005.04.001>
- Li, B., Li, Y., Jiang, W., Su, Z., & Shen, W. (2020). Conjugate ruptures and seismotectonic implications of the 2019 Mindanao earthquake sequence inferred from Sentinel-1 InSAR data. *International Journal of Applied Earth Observation and Geoinformation*. <https://doi.org/10.1016/j.jag.2020.102127>
- Li, G., Wang, A., & Gao, Y. (2023). Source rupture characteristics of the September 5, 2022 Luding M_s 6.8 earthquake at the Xianshuihe fault zone in southwest China. *Earthquake Research Advances*, 3. <https://doi.org/10.1016/j.eqrea.2022.100201>
- Linoff, G. S., & Berry, M. J. (2011). *Data Mining Techniques: For Marketing*. Wiley.
- Ma, K.-F., & Wu, H.-Y. (2001). Quick slip distribution determination of moderate to large inland earthquakes using near-source strong motion waveforms. *Earthquake Engineering and Engineering Seismology*, 3, 1–10.
- Mai, P. M., & Beroza, G. C. (2000). Source scaling properties from finite-fault-rupture models. *Bulletin of the Seismological Society of America*, 90, 604–615. <https://doi.org/10.1785/0119990126>
- Mai, P. M., & Thingbaijam, K. K. S. (2014). SRCMOD: An online database of finite-fault rupture models. *Seismological Research Letters*, 85, 1348–1357. <https://doi.org/10.1785/0220140077>
- Malakar, S., & Rai, A. K. (2022). Seismicity clusters and vulnerability in the Himalayas by machine learning and integrated MCDM models. *Arabian Journal of Geosciences*, 15, 1674. <https://doi.org/10.1007/s12517-022-10946-1>
- Malakar, S., Rai, A. K., & Gupta, A. K. (2022). Earthquake risk mapping in the Himalayas by integrated analytical hierarchy process, entropy with neural network. *Natural Hazards*. <https://doi.org/10.1007/s11069-022-05706-z>
- Mulia, I. E., Gusman, A. R., Heidarzadeh, M., & Satake, K. (2022). Sensitivity of Tsunami data to the up-dip extent of the July 2021 Mw 8.2 Alaska earthquake. *Seismological Research Letters*, 93, 1992–2003. <https://doi.org/10.1785/0220210359>
- Murotani, S., Matsushima, S., Azuma, T., et al. (2015). Scaling relations of source parameters of earthquakes occurring on inland crustal mega-fault systems. *Pure and Applied Geophysics*, 172, 1371–1381. <https://doi.org/10.1007/s00024-014-1010-9>
- Nedic, V., Despotovic, D., Cvetanovic, S., Despotovic, M., & Babic, S. (2014). Comparison of classical statistical methods and artificial neural network in traffic noise prediction. *Environmental Impact Assessment Review*, 49, 24–30. <https://doi.org/10.1016/j.eiar.2014.06.004>
- Nowroozi, A. A. (1985). Empirical relations between magnitudes and fault parameters for earthquakes in Iran. *Bulletin of the Seismological Society of America*, 75, 1327–1338.
- Ohnaka, M. (1978). Earthquake-source parameters related to magnitude. *Geophysical Journal International*, 55, 45–66. <https://doi.org/10.1111/j.1365-246X.1978.tb04747.x>
- Olson, R. S., Urbanowicz, R. J., Andrews, P. C., Lavender, N. C., Kidd, L. C., & Moore, J. H. (2016). Automating biomedical data science through tree-based pipeline optimisation. *Applications of Evolutionary Computation*. https://doi.org/10.1007/978-3-319-31204-0_9
- Omira, R., Vales, D., Marreiros, C., & Carrilho, F. (2015). Large submarine earthquakes that occurred worldwide in a 1-year period (June 2013 to June 2014)—a contribution to the understanding of tsunamigenic potential. *Natural Hazards and Earth System Sciences*, 15, 2183–2200. <https://doi.org/10.5194/nhess-15-2183-2015>
- Papadimitriou, E. E., & Karakostas, V. G. (2006). Earthquake generation in Cyprus revealed by the evolving stress field. *Tectonophysics*, 423, 61–72. <https://doi.org/10.1016/j.tecto.2006.03.014>
- Papadimitriou, E., Karakostas, V., Tranos, M., et al. (2007). Static stress changes associated with normal faulting earthquakes in South Balkan area. *International Journal of Earth Sciences*, 96, 911–924. <https://doi.org/10.1007/s00531-006-0139-x>

- Rai, A. K., Malakar, S., & Goswami, S. (2023). Active source zones and earthquake vulnerability around Sumatra subduction zone. *Journal of Earth System Science*, 132, 66. <https://doi.org/10.1007/s12040-023-02070-9>
- Ren, T., Wang, P., Lin, M., et al. (2020). Classification of tectonic and nontectonic earthquakes by an integrated learning algorithm. *Pure and Applied Geophysics*, 177, 455–467. <https://doi.org/10.1007/s00024-019-02233-0>
- Reyes, J., Morales-Esteban, A., & Martínez-Álvarez, F. (2013). Neural networks to predict earthquakes in Chile. *Applied Soft Computing*, 13, 1314–1328. <https://doi.org/10.1016/j.asoc.2012.10.014>
- Romano, F., Gusman, A. R., Power, W., Piatanesi, A., Volpe, M., Scala, A., & Lorito, S. (2021). Tsunami source of the 2021 MW 8.1 Raoul Island earthquake from DART and tide-gauge data inversion. *Geophysical Research Letters*, 48, e2021GL094449. <https://doi.org/10.1029/2021GL094449>
- Shaw, B. E. (2013). Earthquake surface slip-length data is fit by constant stress drop and is useful for seismic hazard analysis. *Bulletin of the Seismological Society of America*, 103, 876–893. <https://doi.org/10.1785/0120110258>
- Sims, B. H., Wiedlea, K. A., & Wilson, G. D. (2008). Expert opinion in reliability. *Encyclopedia of Statistics in Quality and Reliability*.
- Slemmons, D. B. (1982). Determination of design earthquake magnitudes for microzonation. *Proceeding of the Third International Earthquake Microzonation Conference*, 119–130.
- Slemmons, D. B., Bodin, P., & Zang, X., 1989. Determination of earthquake size from surface faulting events. *Proceeding of the International Seminar on Seismic Zonation, Guangzhou, China*, 13.
- Stirling, M., Goded, T., Berryman, K., & Litchfield, N. (2013). Selection of earthquake scaling relationships for seismic-hazard analysis. *Bulletin of the Seismological Society of America*, 103, 2993–3011. <https://doi.org/10.1785/0120130052>
- Stirling, M., Rhoades, D., & Berryman, K. (2002). Comparison of earthquake scaling relations derived from data of the instrumental and preinstrumental era. *Bulletin of the Seismological Society of America*, 92, 812–830. <https://doi.org/10.1785/0120000221>
- Styron, R., & Pagani, M. (2020). The GEM global active faults database. *Earthquake Spectra*, 36, 160–180. <https://doi.org/10.1177/8755293020944182>
- Szafranski, D., & Duan, B. (2022). A workflow to integrate numerical simulation, machine learning regression and bayesian inversion for induced seismicity study: principles and a case study. *Pure and Applied Geophysics*, 179, 3543–3568. <https://doi.org/10.1007/s00024-022-03140-7>
- Thingbaijam, K., Mai, P. M., & Goda, K. (2017). New empirical earthquakesource scaling laws. *Bulletin of the Seismological Society of America*, 107, 2225–2246. <https://doi.org/10.1785/0120170017>
- Thomas, P., & Suhner, M. C. (2015). A new multilayer perceptron pruning algorithm for classification and regression applications. *Neural Processing Letters*, 42, 437–458. <https://doi.org/10.1007/s11063-014-9366-5>
- Vakov, A. V. (1996). Relationships between earthquake magnitude, source geometry and slip mechanism. *Tectonophysics*, 261, 97–113. [https://doi.org/10.1016/0040-1951\(96\)82672-2](https://doi.org/10.1016/0040-1951(96)82672-2)
- Wagner, S., Kronberger, G., et al. (2014). Architecture and design of the heuristiclab optimization environment. *Advanced Methods Applications Computational Intelligence*. https://doi.org/10.1007/978-3-319-01436-4_10
- Wells, D. L., & Coppersmith, J. C. (1994). New empirical relationships among magnitude, rupture length, rupture width, rupture area and surface displacement. *Bulletin of the Seismological Society of America*, 84, 974–1002. <https://doi.org/10.1785/BSSA0840040974>
- Wesnowsky, S. G. (2008). Displacement and geometrical characteristics of earthquake surface ruptures: Issues and implications for seismic hazard analysis and the process of earthquake rupture. *Bulletin of the Seismological Society of America*, 98, 1609. <https://doi.org/10.1785/0120070111>
- Wong, T.-T. (2015). Performance evaluation of classification algorithms by k-fold and leave-one-out cross validation. *Pattern Recognition*, 48, 2839–2846. <https://doi.org/10.1016/j.patcog.2015.03.009>
- Xu, G., Xu, C., Wen, Y., & Jiang, G. (2017). Source parameters of the 2016–2017 central Italy earthquake sequence from the sentinel-1, ALOS-2 and GPS data. *Remote Sensing*, 9, 1182. <https://doi.org/10.3390/rs9111182>
- Xu, L., Yunjun, Z., Ji, C., Meng, L., Fielding, E. J., Zinke, R., & Bao, H. (2023). Understanding the rupture kinematics and slip model of the 2021 Mw 7.4 Maduo earthquake: A bilateral event on bifurcating faults. *Journal of Geophysical Research: Solid Earth*, 128, e2022JB025936. <https://doi.org/10.1029/2022JB025936>
- Xu, X., Sandwell, D. T., & Smith-Konter, B. (2020). Coseismic displacements and surface fractures from sentinel-1 InSAR: 2019 ridgecrest earthquakes. *Seismological Research Letters*, 91, 1979–1985. <https://doi.org/10.1785/0220190275>
- Zhang, G., Eddy, P. B., & Hu, M. Y. (1998). Forecasting with artificial neural networks: The state of the art. *International Journal of Forecasting*, 14, 35–62. [https://doi.org/10.1016/S0169-2070\(97\)00044-7](https://doi.org/10.1016/S0169-2070(97)00044-7)